

Conjecture 00000000957: frame potential local minima

Conjecture. “Local minima of FP occur at r -fold repetitions of orthogonal bases for frames of redundancy r ; and the classification of minima is complete at integer redundancy.”

Reading. A frame is $N = rd$ unit-norm vectors $\varphi_1, \dots, \varphi_N \in \mathbb{R}^d$ spanning $\mathbb{R}^d$ (lower bound $A\|x\|^2 \leq \sum_i \langle x, \varphi_i \rangle^2$, $A > 0$), $r \geq 1$ the integer redundancy. $\text{FP}(\Phi) = \sum_{i,j} \langle \varphi_i, \varphi_j \rangle^2$, and local minimality is with respect to the topology of the space of unit-norm N -tuples. An r -fold repetition of an orthonormal basis is a tuple with $\varphi_i = s_i e_{f(i)}$, $s_i \in \{\pm 1\}$, (e_k) orthonormal, $|f^{-1}(k)| = r$ for all k (signs allowed, which is more generous than the strict notion, so refuting it refutes the strict one). The “classification is complete” clause is read as: every local minimum is of this form. This is the statement **C957.Conjecture**. Not refuted: the converse (repeated orthonormal bases are local minima, which is true), readings without the unit-norm constraint, complex scalars, non-integer r . No literature is used.

Counterexample. $d = 2$, $r = 3$, $N = 6$: the vectors $u_0 = (1, 0)$, $u_1 = (\frac{1}{2}, \frac{\sqrt{3}}{2})$, $u_2 = (-\frac{1}{2}, \frac{\sqrt{3}}{2})$, each taken twice (the doubled Mercedes-Benz frame).

Lemma 1 (Lean **C957.fp_eq**). *For unit vectors $\varphi_i = (x_i, y_i) \in \mathbb{R}^2$ put $a_i = x_i^2 - y_i^2$, $b_i = 2x_i y_i$. Then $\text{FP} = \frac{N^2}{2} + \frac{1}{2}((\sum a_i)^2 + (\sum b_i)^2)$.*

Proof. Using $x_i^2 + y_i^2 = 1 = x_j^2 + y_j^2$ one has $2(x_i x_j + y_i y_j)^2 = 1 + a_i a_j + b_i b_j$, because the difference equals $(x_i^2 + y_i^2)(x_j^2 + y_j^2) - 1$. Sum over i, j . $\square$

Hence $\text{FP} \geq N^2/2 = 18$ on all unit-norm 6-tuples. For the doubled Mercedes-Benz frame $\sum a_i = 2(1 - \frac{1}{2} - \frac{1}{2}) = 0$ and $\sum b_i = 2(0 + \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2}) = 0$, so $\text{FP} = 18$: a global minimum, hence a local one (**C957.mb_isMin**, **C957.mb_isLocalMin**). It is a frame, indeed $\sum_i \langle x, \varphi_i \rangle^2 = 3\|x\|^2$ (**C957.mb_isFrame**).

It is not a signed repetition of an orthonormal basis (**C957.mb_not_repeated**): if $\varphi_0 = s_0 e_a$, $\varphi_1 = s_1 e_b$ with (e_k) orthonormal, then $\langle \varphi_0, \varphi_1 \rangle = s_0 s_1 \langle e_a, e_b \rangle \in \{0, \pm 1\}$, but $\langle u_0, u_1 \rangle = \frac{1}{2}$.

Lean. File **Conjecture957/Basic.lean** (namespace **C957**); the statement block defines **FP**, **IsFrame**, **IsRepeatedONB**, **Conjecture** and **Claim** := $\neg$ **Conjecture**; the main theorem is **C957.main** : **C957.Claim**. Convention: **inner** is the real Euclidean inner product on EuclideanSpace $\mathbb{R}$ (Fin d). Axiom audit (**#print axioms C957.main**): **propext**, **Classical.choice**, **Quot.sound**; no **sorry**, no **native_decide**.